EChem MT 1 $i_{anode} + i_c = 0$; $i_{anode} > 0$; $\sum z_j c_j = 0$

Basics: $\bar{n} \cdot N_j = -\frac{s_j}{nF} i_n$; $i_n = F \sum z_j \bar{n} \bar{N}_j$; $m_i = \frac{-s_i (MW) Q}{nF}$

$\Delta G_n = -nFU$; U = cell potentials

$\bar{i} = -\kappa \nabla \phi = -\sigma \nabla \phi$

$\kappa = F^2 \sum z_j^2 u_j c_j$

$u_j = \frac{D_j}{RT}$ [Nernst Einstein]

$t_j = \frac{z_j^2 u_j c_j}{\sum_k z_k^2 u_k c_k}$; $\sum t_j = 1$; $t_+ = \frac{z_+ u_+}{z_+ u_+ - z_- u_-}$ (binary)

↳ **Double layer:** $\lambda = \sqrt{\frac{\varepsilon RT}{F^2 \sum_j z_j^2 c_{j\infty}}}$; $I_s = \frac{1}{2} \sum z_j^2 c_{j\infty}$; $\kappa = \frac{\varepsilon}{\varepsilon_0}$; ionic strength

$I_s = \frac{1}{2} \sum_j z_j^2 c_{j\infty}$; $C = \frac{\varepsilon}{\lambda}$; $\frac{1}{C_{dl}} = \frac{1}{C_{GHP}} + \frac{1}{C_{OHP}} + \frac{1}{C_d}$

C_{dl} = minval of diffuse layer capacitance

Kinetics: $W = -F \Delta V$; $\eta_s = V - U$

$i = i_0 \left[\exp\left(\frac{(1-\beta) F \eta}{RT}\right) - \exp\left(\frac{-\beta F \eta}{RT}\right) \right]$; $(1-\beta) = \alpha_a$; $+\beta = \alpha_c$

↳ **Linear:** $i = \frac{i_0 (\alpha_a + \alpha_c) F}{RT} \eta_s$

↳ $\eta \gg 0$: $i = i_0 \exp\left(\frac{\alpha_a F \eta_s}{RT}\right)$; $i = -i_0 \exp\left(\frac{-\alpha_c F \eta_s}{RT}\right)$

↳ **Mechanism:**
- Elementary steps (1 e⁻ at a time)
- Rate law for each step
- Assume RLS, set others to QE → eliminate intermediates
- Compose overall RL
- Determine reversible potential (U), introduce η, find i_0 ; $r_{F-M} = k_{1,a}[\,] \exp\left(\frac{(1-\beta)FV}{RT}\right) - k_{1,c}[\,] \exp\left(\frac{-\beta FV}{RT}\right)$

e⁻ equations → B-V kinetics (w/ β & V)

$r_{net} = \frac{i}{nF} = r_{lim}$; @ $i = 0$, $V = U$, $i_0 = -=-$ → plug in → $i = f(\eta)$

Thermodynamics $\sum_j s_j M_j = ne^-$; $\mu_j = \mu_j^{chem} + z_j F \phi = RT \ln(\lambda_j)$; $\lambda_j = c_j f_j a_j^\circ$

$z_+ \nu_+ + z_- \nu_- = 0$; $C = \frac{c_+}{\nu_+} = \frac{c_-}{\nu_-}$; P

$$(U_i - U^\circ_{H_2/H^+}) = (U_i^\circ - U^\circ_{H_2/H^+}) - \frac{RT}{nF} \left(\sum_j s_j \ln c_{ij}^* \right)$$

; $c_{ij}^* = c_{ij}$ for solute ; $c_{ji}^* = P_{ij}$ for gas

$U_{cell} = V^{\alpha'} - V^{\alpha}$ [1 M aq., 1 bar P]

$\mu_e^{\alpha} - \mu_e^{\alpha'} = F U_{cell}$

$U_{cell} = U^\circ_{cell} + \frac{RT}{F} \ln(f(P,c))$

$P_{O_2} = C_{C_n} = 1$

Transport $\bar{N}_j = -z_j u_j F c_j \nabla \phi - D_j \nabla c_j + c_j \bar{v}$ → $\bar{i} = F \sum_j z_j \bar{N}_j = -F \left(\sum z_j^2 u_j F c_j \nabla \phi + z_j D_j \nabla c_j \right)$ → $\bar{i} = -\kappa \nabla \phi$ (Ohm's law)

($\sum z_j c_j = 0$)

Mass conservation: $\nu \frac{\partial c_j}{\partial t} = -\nabla \cdot \bar{N}_j + R_j$ → @ SS, No R_j :

$$\nabla \cdot \bar{N}_j = -D_j \left(\frac{z_j F}{RT} \nabla \cdot (c_j \nabla \phi) + \nabla^2 c_j \right) + \nabla c_j \cdot \bar{v}$$

(thermo)

BC's: @ insulating : $\bar{n} \cdot \bar{N}_j = 0$ or $\bar{n} \cdot \bar{i} = 0$

@ electrode surface : $\bar{n} \cdot \bar{N}_j = \frac{-s_j}{nF} i_n$; $i_n = i_0 \left[\exp\left(\frac{\alpha_a F \eta_s}{RT}\right) - \exp\left(\frac{-\alpha_c F \eta_s}{RT}\right) \right]$; $\eta = V - \phi - U$ (↳ variable)

Limitations for BC's:
1) MT (limiting current density): $c_j = 0$ @ surface
2) Rxn limited (ohmically limited): $c_j^{surf} = c_j^{bulk}$; $\nabla^2 \phi = 0$
3) Very slow charge transfer: uniform current density: $i_n = \frac{I}{\text{area}}$

$\frac{c_+}{\nu_+} = \frac{c_-}{\nu_-} = c$; $z_+ \nu_+ + z_- \nu_- = 0$

Limiting cases for governing eqs:

1) Uniform Composition (well-mixed, no c-gradients): $\frac{i_{avg}}{i_{lim}} \ll 1$ → $\frac{d^2\phi}{dx^2} = 0$

@ $x = 0$: $-\kappa \frac{d\phi}{dx} = i_{avg} = -i_0 \exp\left(-\frac{\alpha_c}{RT} F (V - \phi)\right)$

@ $x = L$: $\phi = 0$

$\therefore \phi = -\frac{i_{avg}}{\kappa}(x - L)$; $i_{avg} = -i_0 \exp\left(\frac{-\alpha_c F (V - i_{avg} L/\kappa)}{RT}\right)$

2) Limiting current density (reactant goes to zero at electrode): $c_j^{lim} = 0$ @ electrode surface, $t_{lim} = \bar{v} = 0$; SS

$\nabla^2 c_{lim} = 0$, c_{lim} @ $x = 0$ ⇒ $c_{lim} = c_{bulk}$

$i_{lim} = \frac{nFD}{s} \left. \frac{\partial c_{lim}}{\partial x} \right|_{x=0}$

$i_{lim} = \frac{nFD}{s} \frac{c_{bulk}}{s}$

R_{ct}, R_Ω

Current distributions

$\frac{i_{avg}}{i_{lim}} = \begin{cases} 0 & \text{[uniform comp]} \ \nabla c = 0 \\ 1 & \text{[MT limitations]} \end{cases}$; ∇c is imp

$W_a = \frac{R_{ct}}{R_\Omega}$ = charge transfer Resistance / Ohmic resistance } related to electrolyte

$W_a = \begin{cases} <0.1 & \text{[Maximum non-uniformity]} \\ >0.1 & \text{[uniform current dist.} \rightarrow \text{no MT resistances]} \end{cases}$

Tertiary current dist.

$N_j = -z_j u_j F c_j \nabla\phi - D_j \nabla c_j + v c_j$; $\sum z_j c_j = 0$

→ Can ∇c_j be neglected? ($\frac{i_{avg}}{i_{lim}} = ?$)

No ↓ ~1: Ignore E-field ($t_j = 0$) → $\frac{\partial c_j}{\partial t} + \bar{v}\cdot\nabla c_j = D_j \nabla^2 c_j$ [Convection-Diffusion eq]

~0: $\nabla\phi^2 = 0$ [Laplace eq]

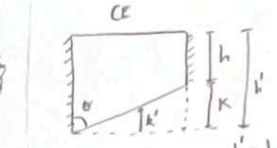
R_{ct}? → Secondary Current Dist. ($W_a > 0$)

N ($W_a \to 0$) → Primary Current Dist.

Primary Current Dist.
1) $\nabla c_j = 0$
2) R_Ω dominates
3) Const. potential BC's
4) Max non-uniformity

Secondary Current Dist.
1) $\nabla c_j = 0$
2) R_Ω, R_{ct}
3) (i-n) type BC's [tafel, linear]
4) More uniform than primary

$-\kappa \frac{\partial\phi}{\partial y} = i_0 \exp\left(\frac{\alpha_a F (V-\phi)}{RT}\right)$



$h' = h + (k - k')$; $k = \frac{L}{\tan\theta}$, $k' = \frac{x}{\tan\theta}$

Hemispherical Electrode ($\frac{i_{avg}}{i_{lim}} \ll 1$)

$\nabla^2\phi = 0$; BC's: $\phi = V$ @ $r = r_0$; $\phi = 0$ @ $r = \infty$



$\phi = f(V, r, r_0)$

$i_\Omega = -\kappa \frac{\partial\phi}{\partial r}$; $R_\Omega = \frac{V}{i_\Omega}$

$R_{ct} = \frac{\partial\eta}{\partial i}\Big|_{i_{avg}}$; $W_a = \frac{R_{ct}}{R_\Omega}$

Linear ↓ → Tafel → BV

Hull cell ($\frac{i_{avg}}{i_{lim}} \ll 1$)

$\nabla^2\phi = 0$; BC's: @ $y = h'$, $\phi = 0$

@ $y = k - k'$, $\phi = V$

@ $x = 0$ $\frac{\partial\phi}{\partial x} = 0$ } insulating BC

@ $x = L$ $\frac{\partial\phi}{\partial x} = 0$ }

$\frac{i(x)}{i_{avg}} = \frac{\left(\frac{x}{L}\right)^a}{\left(1 - \frac{x}{L}\right)^b}\left(c - d\frac{k}{L}\right)$

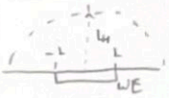
Line Electrode ($\frac{i_{avg}}{i_{lim}} \ll 1$)

$\frac{\partial^2\phi}{\partial x^2}, \frac{\partial^2\phi}{\partial y^2} = 0$

$\frac{\partial\bar{\phi}}{\partial\bar{y}} = -\frac{(1-\bar{\phi})}{W_{a_L}}$ [Linear]

$\frac{\partial\bar{\phi}}{\partial\bar{y}} = -A\exp(-\bar{\phi})$

$A = \frac{1}{W_{a_T}}$



BC's: 1) $x^2 + y^2 = L_w^2$; $\phi = 0$

2) $|x| \geq L$; $\frac{\partial\phi}{\partial y} = 0$

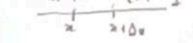
3) $|x| < L$; $\phi = V$ } primary

@ $y = 0$

4) $|x| < L$ } secondary

@ $y = 0$

Electrode conductivity $\eta = \phi_m - \phi_s$



$i_x = -a\frac{di_m}{dx}$; $i_y = \frac{d}{dx}\left(\frac{1}{R_s}\frac{d\phi_m}{dx}\right)$

$i_m = -\frac{1}{aR_s}\frac{d\phi_m}{dx}$; $\bar{K}_0 = \frac{kL}{a\sigma_a}$



BC's: @ $x = L$, $\frac{d\phi_m}{dx} = 0$ (Insulator)

@ $x = 0$, $\phi_m = V$

$\frac{W_a}{\bar{K}_0}$ → Large } more uniform ; $\bar{K}$ ↑ σ ↑ } More uniform

RDE's:

momentum BL

$\xi = z/\sqrt{\frac{\nu}{\Omega}}$; $-a\bar{z}^2\frac{dc}{d\bar{z}} = \frac{d^2c}{d\bar{z}^2}$

$g = \frac{d_c}{d\bar{z}}\left(\frac{1}{2} - a\bar{z}^2 g\right) = \frac{dg}{d\bar{z}}$ → $\frac{dc}{dz}$ → $N = -D\frac{dc}{dz}\Big|_{z=0}$ ≈ $N = \frac{DC_\infty}{\delta}$ →

BC's: @ $\bar{z} = 0$, $c = 0$; @ $\bar{z} = \infty$, $c = c_\infty$

$\bar{z} = \left(\frac{\Omega^{1/2}}{\nu^{1/2} D^{1/3}}\right) z$; $t_j = \frac{z_j^2 u_j c_j}{\sum_k z_k^2 u_k c_k}$; Ratio of mobilities of ions; $t_j \to 0$ (electrical migration can be neglected)

Faraday's Law

collection efficiency (fraction of species reacted @ ring that @ disk) $f(r_1, r_2, r_3)$

RRDE:

+ Generator-Collector

↳ $c_{ring} = 0$, $i_{ring} = \frac{nFD}{\delta}\int_{r_2}^{r_3}\frac{\partial c}{\partial y}\Big|_{y=0} r\,dr$; $i_{ring} = -N_c i_{disk}$ } Ring potential is held s.t. generated species conc is zero

+ Detection mode

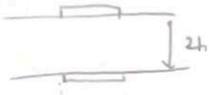
↳ $(c_{ring} - c_\infty) = N_0(c_{disk} - c_\infty)$

↳ detection efficiency $f(r_1, r_2, r_3)$

(fraction of species generated @ disk, detected as surface @ ring)

Microfluidic Ecell

($\frac{i_{avg}}{i_{lim}} = 1$) MT limit



Convection-Diff:

$v\frac{\partial c}{\partial x} = D\left(\frac{\partial^2 c}{\partial x^2} + \frac{\partial^2 c}{\partial y^2}\right)$

$P >> 1$, convective dominated (ignore x-direction diff)